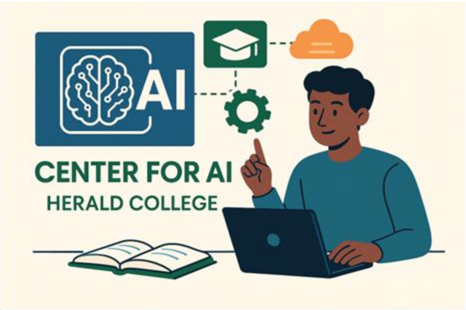
HCAI5TML01 – Mathematics of Learning.

Theme: Towards VC Generalization Bound.

Lec01 – From Averages to Guarantees: Hoeffding Inequality to PAC Framework.

Siman Giri



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Our Objective …

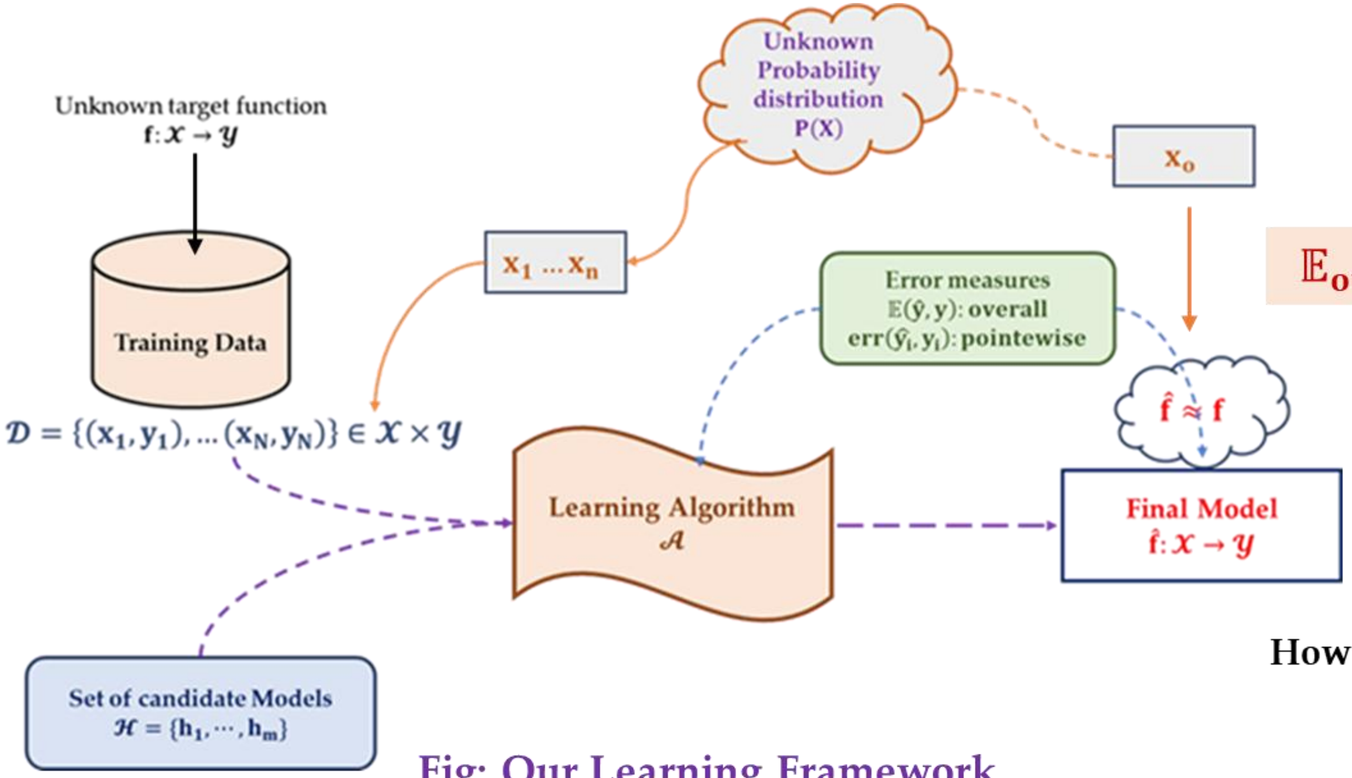


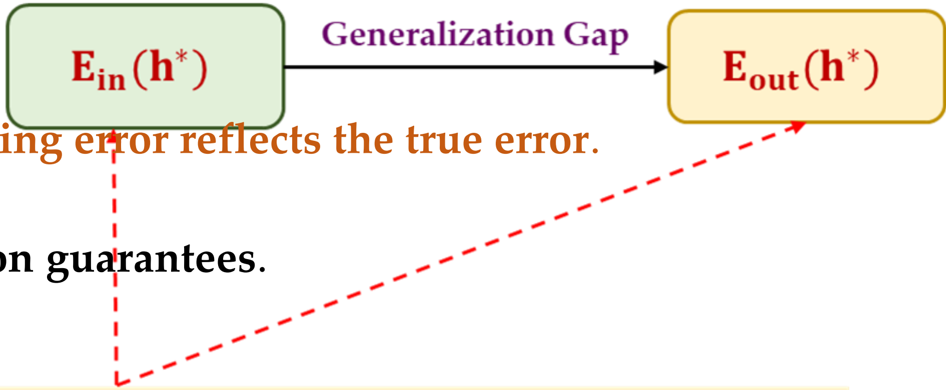
Fig: Our Learning Framework.

$$\mathbb{E}_{out}(\mathbf{h}) = \mathbb{P}[\mathbf{h}(\mathbf{x}) \neq \mathbf{f}(\mathbf{x})]$$

How do we compute such probability?

Why we care about Inequalities in Learning Theory?

- In **General learning** we want to bound the **out – of – sample error**:
  - $\mathbb{E}_{out}(h^*) = \mathbb{P}[h(x) \neq f(x)]$
    - **Problem:**
      - The **true distribution of x** is unknown, so we **cannot compute this probability directly**.
    - **Solution:**
      - Use **probability inequalities** to bound how far the **in-sample error $E_{in}(h^*)$** can be from **$E_{out}(h^*)$**
    - **Intuition:**
      - Even if we do not know **the exact distribution**,
        - inequalities give **guarantees on how likely the training error reflects the true error**.
    - **Result:**
      - This is the **foundation for PAC bounds and generalization guarantees**.



- **Probability Bounds:**
  - **Hoeffding/Chernoff:** $\mathbb{P}(|E_{in} - E_{out}| \geq \epsilon) \leq 2Me^{-2\epsilon^2 N}$
  - **Chebyshev:** $\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$
  - **Markov:** $\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$

Fig: Inequalities and Probabilistic Bound for Learning.

1. Hoeffding's Inequality.

1.1 Hoeffding's In – Equality: The Idea.

- **Hoeffding's inequality** gives a **bound on the probability:**
  - that the sum of independent bounded random variables **deviates** from its expected value.
- It's **particularly** useful when:
  - You know **each variable's min and max** values (boundedness)
  - The variables are **independent** but not necessarily identically distributed

Hoeffding's Inequality - Statement:

Let $\mathbf{X_1, X_2, \dots, X_n}$ be independent random variables such that:

$$\mathbf{a_i \leq X_i \leq b_i} \quad \text{for all } \mathbf{i = 1, \dots, n}$$

Then for any $\mathbf{t > 0}$:

$$\mathbf{P}\left(\frac{1}{n}\sum_{i=1}^{n} \mathbf{X_i} - \mathbf{E}\left[\frac{1}{n}\sum_{i=1}^{n}\mathbf{X_i}\right] \geq \mathbf{t}\right) \leq \exp\left(-\frac{\mathbf{2n^2t^2}}{\sum_{i=1}^{n}(\mathbf{b_i - a_i})^2}\right)$$

where:

- $\mathbf{t}$: Deviation threshold from the mean
- $\mathbf{b_i - a_i}$: Range of each random variable
- $\mathbf{n}$: Number of samples
- The bound decays **exponentially** with $\mathbf{t^2}$

Interpretation:

- The inequality bounds the probability that the sample mean deviates from its expected value
- The bound decays exponentially with $\mathbf{t^2}$
- Tighter when the ranges $(\mathbf{b_i - a_i})$ are small
- Applies to any bounded random variables (not necessarily identically distributed)

1.2 What are a_i and b_i?

- In **Hoeffding's Inequality**,
  - the values a_i and b_i represent the **lower and upper bounds**, respectively,
    - on the possible values of the random variable X_i.
  - **where:**
    - $\mathbf{a_i = \text{Minimum possible value of } X_i}$
    - $\mathbf{b_i = \text{Maximum possible value of } X_i}$
- **Where do we get a_i and b_i?**
  - These **bounds** are derived from **the problem setup or known properties of the random variables**:
    - **Bounded distributions:**
      - If X_i is known to lie in a **fixed range (e.g. $X_i \in [0, 1]$),** then:
        - $\mathbf{a_i = 0}$ & $\mathbf{b_i = 0}$
    - **Physical constraints:**
      - If X_i represents measurement (e.g. temperature, test scores), the bounds come from **domain knowledge**:
        - E.g. If X_i is **human height (in cm)** $\Rightarrow \mathbf{a_i = 50, b_i = 250}$
    - **Theoretical guarantees:**
      - Some algorithms (e.g. **in machine learning**) produce **output with known bounds**.

1.2.1 Why are a_i and b_i important?

1. **Tighter bounds:**
   - The smaller $(\mathbf{b_i} - \mathbf{a_i})$, the **stronger the inequality**.
2. **Exponential decay:**
   - The probability of deviation **decreases exponentially with $\mathbf{t^2}$** and **inversely** with $\sum(\mathbf{b_i} - \mathbf{a_i})^2$

- **Example:**

Given

Let $X_1, \ldots, X_n$ be independent random variables where each X_i is bounded:

$$0 \leq X_i \leq 1 \quad \text{(i.e., } a_i = 0, b_i = 1\text{)}$$

Hoeffding's Inequality Simplifies to

$$P\left(\frac{1}{n}\sum_{i=1}^{n} X_i - E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] \geq t\right) \leq e^{-2nt^2}$$

Why?

1. Original denominator: $\sum_{i=1}^{n}(b_i - a_i)^2 = \sum_{i=1}^{n}(1-0)^2 = n$
2. Thus, the bound becomes:

$$\exp\left(-\frac{2n^2t^2}{n}\right) = e^{-2nt^2}$$

Interpretation

- **Tighter Bound**: The probability of deviation decays **exponentially** with n and t^2
- **Practical Use**: If $n = 100$ and $t = 0.1$, the bound is $e^{-2\times 100\times 0.01} = e^{-2} \approx 0.135$
- **Comparison**: For $t = 0.1$, Chebyshev's bound would give $\frac{\text{Var}}{nt^2}$ (much looser for small t)

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- **Example:**

Why the Simplification Occurs

For variables $X_i \in [0,1]$:

$$(b_i - a_i) = 1 \quad \text{for all } i \implies \sum_{i=1}^{n}(b_i - a_i)^2 = n$$

Thus, the general form:

$$\exp\left(-\frac{2n^2t^2}{\sum(b_i - a_i)^2}\right) \quad \text{reduces to} \quad e^{-2nt^2}$$

Exponential Decay Significance

The bound e^{-2nt^2} has crucial properties:

- **Stronger than Chebyshev**: Polynomial decay $(\frac{1}{t^2})$ vs exponential decay (e^{-t^2})
- **Preferred for Bounded Variables**: Explains why Hoeffding is widely used in ML for bounded losses

Dependence on Parameters

Parameter	Effect on Bound
Sample size $n \uparrow$	Tighter concentration ($\downarrow$ probability)
Deviation $t \uparrow$	Exponentially smaller probabilities

Implications

- **Boundedness is Crucial**: The $[0,1]$ constraint enables the strong exponential bound
- **Practical Impact**: For $n = 1000$, $P(\text{deviation} \geq 0.05) \leq e^{-5} \approx 0.0067$

1.2.2 What is t or ϵ?

- In the context of **Hoeffding's Inequality**,
  - the **variable t** represents the **deviation threshold** or **margin of error**
    - for **the sample mean** from its **expected value.**
- **Definition of t:**
  - **t** is a **positive real number** $(\mathbf{t} > \mathbf{0})$ that quantifies:
    - How far the observed sample mean $\frac{1}{n}\sum_{i=1}^{n} X_i$ is allowed
      - to deviate from the true expected value $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$.
    - The "*tolerance level*" for the deviation.

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Then for any $\mathbf{t} > \mathbf{0}$:

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1.3 Example: Hoeffding Inequality in Action.

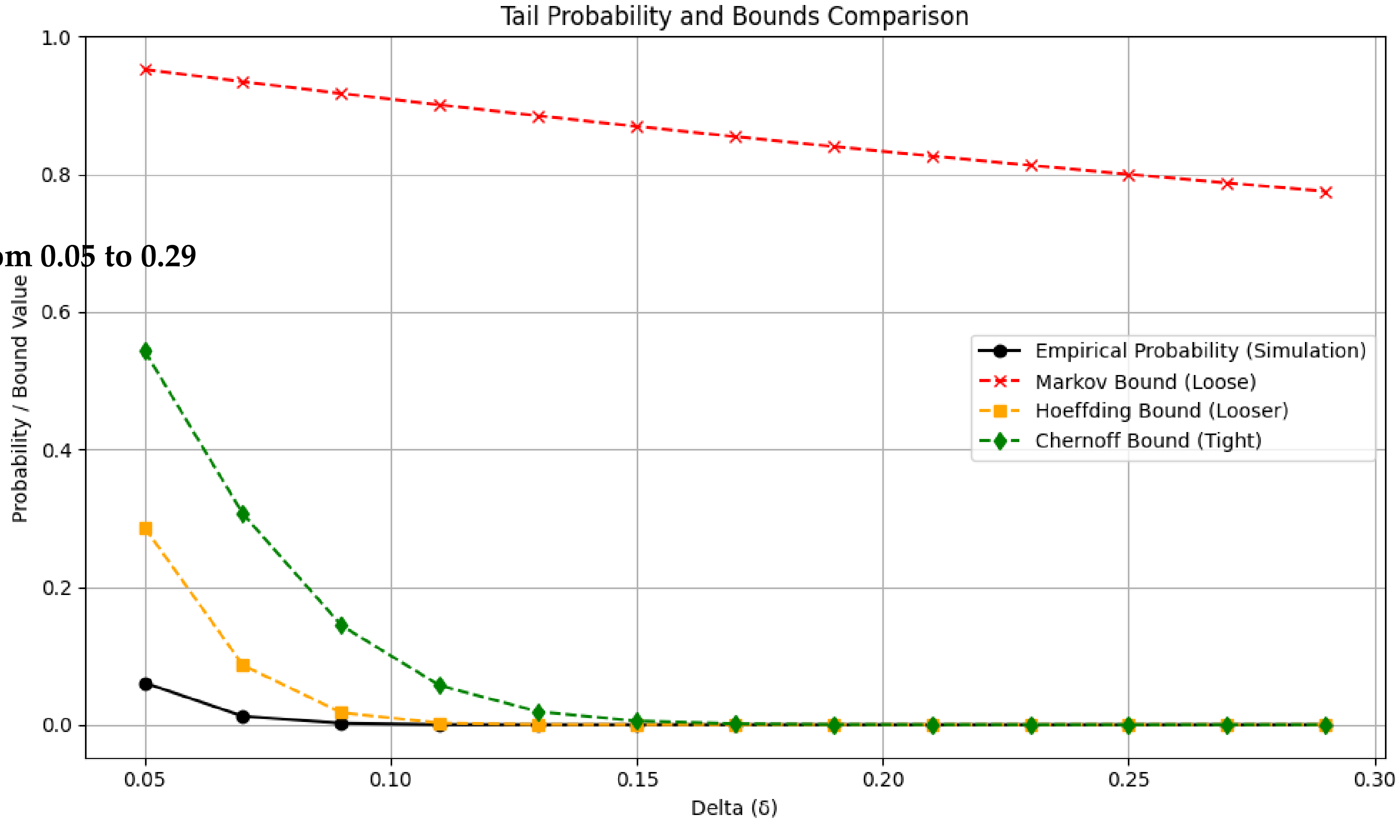
- **Scenario:**
  - We flip a fair coin $\mathbf{n = 1000}$ times.
  - The expected number of heads is:
    - $\boldsymbol{\mu = n \cdot p = 1000 \cdot 0.5 = 500}$
  - We ask:
    - **What is the probability of getting at least 550 heads?**
      - {That's **50 more heads** than expected.}

1.3.1 Example: Hoeffding Inequality in Action.

- **Step 1 – Parameters:**
  - $t = 550 - 50 = 50$
  - **Hoeffding (one-sided bound for independent variables in [0,1]):**
    - $P(X - \mu \geq t) \leq \exp\left(-\frac{2t^2}{n}\right)$
- **Step 2 - Apply Hoeffding:**
  - $P(X \geq 550) \leq \exp\left(-\frac{2 \cdot 50^2}{1000}\right)$
  - $P(X \geq 550) \leq \exp\left(-\frac{5000}{1000}\right) = \exp(-5) \approx 0.0067$
  - **So Hoeffding tells us:**
    - **There's less than a 0.67% chance** of **getting 550 or more heads out of 1000 flips**.
- **Comparison to Chernoff:**
  - **Chernoff** for same case gave about **9.2% earlier because we used δ=0.1**
    - but with its particular form it was **looser** here.
  - **Hoeffding** is very sharp for **bounded variables** like coin flips
    - because it only depends **on the range [0,1], not the distribution shape.**

1.4 Visualization: Empirical vs Chernoff vs. Hoeffding.

- Experiment parameter:
  - n = 1000 # number of trials
  - p = 0.5 # success probability
  - mu = n * p # expected sum
  - trials = 10000 # number of experiments
  - delta values = [i/100 for i in range(5, 30, 2)] # delta from 0.05 to 0.29



1.5 What makes Hoeffding Special?

1. **Tight Bounds (Sharpness):**
   - **Hoeffding** is derived using **Chernoff's technique**, so it **inherits the exponential concentration properties**.
   - It produces **much sharper (tighter) tail bounds** than **Chebyshev for bounded random variables**.
2. **Distribution – Free (No mean or Variance Needed):**
   - Unlike **Chebyshev**, **Hoeffding** does not require knowledge of $\mathbb{E}[X]$ **or** $\mathrm{Var}[X]$ to **give a bound**.
   - All it needs is **that the random variables are independent and bounded within known ranges**.
3. **Strong Tail Control:**
   - Provides an **exponentially decaying bound** for the probability that **the sum or average deviates** from its **expected value**:
     - $P(|\bar{X} - \mu| \geq \epsilon) \leq 2\exp(-2\epsilon^2 N)$
   - This makes it very useful in learning theory:
     - where we care about **in – sample error approximating out – of - sample error**.

1.6 Comparing Hoeffding and Chebyshev

1. **Hoeffding and Chebyshev Definition:**
   - Both **inequalities give probabilistic guarantees** about how far **the sample mean** (**or in – sample error**) can **deviate** from **the true mean** (**or out – of – sample error**).
     - **Chebyshev Inequality:**
       - $P\left(\left|\frac{1}{N}\sum_{n=1}^{N} X_n - \mu\right| > \epsilon\right) \le \frac{\sigma^2}{N\epsilon^2}$
       - **Here:**
         - ϵ = the **magnitude of deviation** from **the mean μ** you care about(**deviation threshold**).
         - Example: If $\epsilon = 0.1$; you are asking:
           - *"What is **the probability** that **the sample mean is at least 0.1 units** away from **the true mean**?"*
     - **Hoeffding Inequality:**
       - $P\left(\left|\frac{1}{N}\sum_{n=1}^{N} X_n - \mu\right| > \epsilon\right) \le 2\exp\left(\frac{-2N^2\epsilon^2}{\sum_{n=1}^{N}(b_n - a_n)^2}\right)$
       - **Here:**
         - ϵ = the **magnitude of deviation** from **the mean μ** you care about(**deviation threshold**).
         - But now **the bound is exponentially decaying in n and ϵ^2**,
           - making it **much tighter than Chebyshev for bounded variables**.

1.6.1 Comparing Hoeffding and Chebyshev

2. **Sample Size Comparison:**
   - If you want the **deviation probability** to be below a **confidence level δ**:
     - **Solve for N in Hoeffding:**
       - $\mathbf{2\exp\left(-2\epsilon^2 N\right) \leq \delta \Rightarrow N \geq -\frac{1}{2\epsilon^2}\log\frac{\delta}{2}}$
     - **Solve for N in Chebyshev:**
       - $\mathbf{\frac{\sigma^2}{N\epsilon^2} \leq \delta \Rightarrow N \geq \frac{\sigma^2}{\epsilon^2\delta}}$
   - **Example:**
     - **Given:** $\mathbf{\rightarrow \sigma = 1; \epsilon = 0.1, \delta = 0.01}$

Inequality	Required N
Hoeffding	265
Chebyshev	$10,000$

- **Observation:** Hoeffding predicts **far fewer samples** to achieve the same confidence:
  - it is **much tighter for bounded variables**.

1.6.2 Comparing Hoeffding and Chebyshev

3. **Implications for Learning Theory:**
   - In **learning theory**, we often care about **how well the training error predicts the testing error**:
     - $\mathbf{E_{in}(h) = in - sample\ error, E_{out}(h) = out - of - sample\ error}$
   - **Using Hoeffding:**
     - $\mathbf{P(|E_{in}(h) - E_{out}(h)| > \epsilon) \leq 2\exp -2\epsilon^2 N}$
   - As **N (training sample size)** increases:
     - $\mathbf{|E_{in} - E_{out}|}$ becomes **small with high probability**.
       - The **in – sample error** ($\mathbf{E_{in}}$) is something we can **compute numerically using the training set**.
       - The **out – sample error** ($\mathbf{E_{out}}$) is an **unknown quantity** because **we do not know the target function f**.
     - **Hoeffding inequality** says:
       - Even without knowing **the true function f**, for **large enough N** we can **trust**
         - the **training error** as a **good approximation** of **testing error**.
       - Therefore, we will be able to tell **how good the hypothesis function** is **without accessing the unknown target function.**

- **Key insight:**
  - Hoeffding gives us a **quantitative handle on generalization**,
    - it **tells** us **how many samples we need** to **confidently estimate** out – of – sample performance from in – sample performance.

2. PAC Framework.

{Probably Approximately Correct}

2.1 The PAC Framework

- The probabilistic analysis is called a **probably approximately correct (PAC) framework**.
  - The word **P – A – C** comes from **the principles of the Hoeffding inequality**:
    - **Probably:**
      - We use the probability:
        - $\mathbb{P}[|\mathbf{E_{in}}(\mathbf{h}) - \mathbf{E_{out}}(\mathbf{h})|] > \epsilon] \leq 2e^{-2\epsilon^2 N}$
        - as a measure to **quantify the error**.
    - **Approximately:**
      - The **in – sample error** is an **approximation** of **the out – sample error**, as given by:
        - $\mathbb{P}[|\mathbf{E_{in}}(\mathbf{h}) - \mathbb{E}_{\mathbf{out}}(\mathbf{h})| > \epsilon] \leq 2e^{-2\epsilon^2 N}$
        - The **approximation error** is controlled **by ϵ**.
    - **Correct:**
      - The error is **bounded by the right-hand side of the Hoeffding inequality**:
        - $\mathbb{P}[|\mathbf{E_{in}}(\mathbf{h}) - \mathbf{E_{out}}(\mathbf{h})| > \epsilon] \leq 2e^{-2\epsilon^2 N}$
      - The **accuracy is controlled by** N for a **fixed ϵ.**

2.2 Why Hoeffding "as-is" fails for data-dependent g?

- The above **Hoeffding inequality** holds for a **fixed hypothesis function h.**
  - This means that **h is already chosen before we generate the dataset**.
  - If we allow **h** to **change** after **we have generated the dataset**,
    - then **the Hoeffding inequality** is **no longer valid.**
- **What do we mean by after generating the dataset?**
  - In any **learning scenario**, we are given a **training dataset D.**
    - Based on **this dataset**,
      - we have to **choose a hypothesis function g from the hypothesis set H**.
    - The **hypothesis g** we choose depends on **what samples are inside D**
      - and **which learning algorithm A** we use.
    - So, **g changes after the dataset is generated.**
      - **Thus, Hoeffding inequality is invalid if we use g instead h.**

2.3 Make it Valid …

- The final **hypothesis g** is one of **these** $\mathcal{H} \in \{\mathbf{h_1}, \mathbf{h_2}, \dots, \mathbf{g}, \dots, \mathbf{h_m}\}$ **potential hypotheses**.
- To have $|\mathbf{E_{in}(g)} - \mathbf{E_{out}(g)}| > \epsilon$,
  - we need to ensure that **at least one of the M potential hypothesis** can **satisfy the inequality**.
- **This implies:**
  - $$|\mathbf{E_{in}(g)} - \mathbf{E_{out}(g)}| > \epsilon \Rightarrow \begin{array}{c} |\mathbf{E_{in}(h_1)} - \mathbf{E_{out}(h_1)}| > \epsilon \\ \textbf{or}, |\mathbf{E_{in}(h_2)} - \mathbf{E_{out}(h_2)}| > \epsilon \\ \dots \\ \textbf{or}, |\mathbf{E_{in}(h_M)} - \mathbf{E_{out}(h_M)}| > \epsilon \end{array}$$
- If **H is finite …**

2.3.1 Union bound over a finite H

Union Bound over a Finite Hypothesis Class

Let $\mathbf{H} = \{\mathbf{h_1}, \ldots, \mathbf{h_M}\}$. If

$$|\mathbf{E}_{\text{in}}(\mathbf{g}) - \mathbf{E}_{\text{out}}(\mathbf{g})| > \varepsilon,$$

then at least one $\mathbf{h_m} \in \mathbf{H}$ must also violate the margin:

$$|\mathbf{E}_{\text{in}}(\mathbf{h_m}) - \mathbf{E}_{\text{out}}(\mathbf{h_m})| > \varepsilon.$$

Thus,

$$\Pr\left(|\mathbf{E}_{\text{in}}(\mathbf{g}) - \mathbf{E}_{\text{out}}(\mathbf{g})| > \varepsilon\right) \leq \Pr\left(\bigcup_{\mathbf{m=1}}^{\mathbf{M}} \{|\mathbf{E}_{\text{in}}(\mathbf{h_m}) - \mathbf{E}_{\text{out}}(\mathbf{h_m})| > \varepsilon\}\right) \leq \sum_{\mathbf{m=1}}^{\mathbf{M}} \Pr\left(|\mathbf{E}_{\text{in}}(\mathbf{h_m}) - \mathbf{E}_{\text{out}}(\mathbf{h_m})| > \varepsilon\right).$$

Applying Hoeffding to each $\mathbf{h_m}$:

$$\Pr\left(|\mathbf{E}_{\text{in}}(\mathbf{g}) - \mathbf{E}_{\text{out}}(\mathbf{g})| > \varepsilon\right) \leq \sum_{\mathbf{m=1}}^{\mathbf{M}} \mathbf{2e^{-2\varepsilon^2 N}} = \mathbf{2Me^{-2\varepsilon^2 N}}.$$

Interpretation: Selecting $\mathbf{g}$ after seeing data costs a factor of $\mathbf{M}$ (a multiple-comparisons penalty).

2.4 Sample Complexity: Definition.

- Sample complexity refers to **the minimum number of training examples N** required for a **learning algorithm**
  - to **achieve** a **desired level of generalization performance**, typically expressed in terms of:
    - **Accuracy (ϵ):**
      - **How close the in – sample error $E_{in}(h)$ should be to the true out – of – sample error $E_{out}(h)$?**
    - **Confidence ($1 - \delta$):**
      - The probability that bound $|E_{in}(h) - E_{out}(h)| \leq \epsilon$ holds.
- **Mathematical Formulation:**

2.4 Sample Complexity: Definition.

- **Mathematical Formulation:**

Sample Complexity for Finite Hypothesis Class H

Let $\mathbf{H} = \{\mathbf{h_1}, \ldots, \mathbf{h_M}\}$ be a finite hypothesis class. To guarantee with probability at least $\mathbf{1} - \delta$ that

$$|\mathbf{E}_{\text{in}}(\mathbf{g}) - \mathbf{E}_{\text{out}}(\mathbf{g})| \leq \varepsilon,$$

we solve the inequality derived from the union bound and Hoeffding's inequality:

$$\mathbf{2Me}^{-\mathbf{2}\varepsilon^{\mathbf{2}}\mathbf{N}} \leq \delta.$$

Solving for $\mathbf{N}$ (the required number of samples), we get:

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{2}\varepsilon^{\mathbf{2}}}\left(\ln(\mathbf{2M}) + \ln\left(\frac{\mathbf{1}}{\delta}\right)\right).$$

Conclusion:

- **Definition of δ:**
  - δ is the *failure probability* or *tolerance*, representing the maximum allowed probability that the generalization error bound fails to hold (i.e., $|\mathbf{E}_{\text{in}}(\mathbf{g}) - \mathbf{E}_{\text{out}}(\mathbf{g})| > \varepsilon$).
- **Interpretation:**
  - Larger $\mathbf{M}$ (more hypotheses) $\Rightarrow$ requires more data ($\mathbf{N}$) to control overfitting.
  - Smaller δ (stricter confidence) $\Rightarrow$ increases the sample complexity.
  - Tighter ε (error tolerance) $\Rightarrow$ quadratically increases $\mathbf{N}$.

2.4.1 Stronger Guarantees requires More Evidence.

Why Smaller δ Demands More Data?

For a finite hypothesis class **H** of size **M**, the sample complexity **N** scales as:

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{2}\varepsilon^{\mathbf{2}}}\left(\ln(\mathbf{2M}) + \ln\left(\frac{\mathbf{1}}{\delta}\right)\right).$$

- Smaller δ (e.g., $0.01 \to 0.001$) $\Rightarrow$
- Larger $\ln(1/\delta)$ (e.g., $4.6 \to 6.9$).
- Result: N must increase to maintain the same ε.

Intuition: δ is the "risk budget." Lower risk = higher cost in data.

2.4.2 Stronger Guarantees requires More Evidence.

Sample Complexity Example

Example: For fixed $\mathbf{M} = \mathbf{100}$ and $\varepsilon = \mathbf{0.1}$:

- When $\delta = \mathbf{0.05}$ (95% confidence):

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{2} \times \mathbf{0.1^2}} (\ln(\mathbf{200}) + \ln(\mathbf{20})) \approx \mathbf{185}$$

- When $\delta = \mathbf{0.01}$ (99% confidence):

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{0.02}} (\ln(\mathbf{200}) + \ln(\mathbf{100})) \approx \mathbf{265}$$

Key Takeaway:

- $\delta \downarrow \Rightarrow \mathbf{N} \uparrow$
- *Trading confidence for data:* To be more certain that $\mathbf{E}_{\text{out}}$ is close to $\mathbf{E}_{\text{in}}$, you need more samples.

2.5 Sample Complexity for Finite H.

Sample Complexity for Finite Hypothesis Classes

Definition: The minimum number of samples N needed to ensure, with probability $\geq 1-\delta$, that:

$$|\mathbf{E}_{\text{in}}(\mathbf{h}) - \mathbf{E}_{\text{out}}(\mathbf{h})| \leq \varepsilon \quad \forall \mathbf{h} \in \mathbf{H}.$$

Formula:

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{2}\varepsilon^{\mathbf{2}}}\left(\ln(\mathbf{2M}) + \ln\left(\frac{\mathbf{1}}{\delta}\right)\right).$$

Or:

$$\mathbf{N} = \mathbf{\Omega}\left(\frac{\mathbf{1}}{\varepsilon^{\mathbf{2}}}\left(\ln|\mathbf{H}| + \ln\frac{\mathbf{1}}{\delta}\right)\right).$$

Key Takeaways:

Dependencies in Sample Complexity

Dependence on M (Model Complexity)

- **Larger M** (more hypotheses) $\Rightarrow$ **Higher N** needed to avoid overfitting.
- **Example:** Doubling **M** adds only a $\ln(\mathbf{2})$ term to **N**.

Dependence on ε (Precision)

- **Tighter** ε (e.g., $\mathbf{0.1} \to \mathbf{0.01}$) $\Rightarrow$ **N** scales as $\frac{\mathbf{1}}{\varepsilon^{\mathbf{2}}}$.
- **Example:** Halving ε quadruples **N**.

Dependence on δ (Confidence)

- **Smaller** δ (e.g., $\mathbf{0.05} \to \mathbf{0.01}$) $\Rightarrow$ Increases $\ln(1/\delta)$, requiring more data.
- **Example:** Changing δ from 5% to 1% increases **N** by $\ln(20) \approx 3$.

2.5.1 Sample Complexity for Finite H.

Example Calculation of Sample Complexity

Given:

- $\mathbf{M} = \mathbf{100}$ (size of hypothesis class)
- $\varepsilon = \mathbf{0.1}$ (accuracy bound)
- $\delta = \mathbf{0.05}$ (95% confidence level)

The sample complexity is calculated as:

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{2}\varepsilon^{\mathbf{2}}}\left(\ln(\mathbf{2M}) + \ln\left(\frac{\mathbf{1}}{\delta}\right)\right)$$

Plugging in the values:

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{2} \times \mathbf{0.1}^{\mathbf{2}}}(\ln(\mathbf{200}) + \ln(\mathbf{20}))$$

$$\mathbf{N} \geq \frac{\mathbf{1}}{\mathbf{0.02}}(\mathbf{5.298} + \mathbf{2.996})$$

$$\mathbf{N} \geq \mathbf{50} \times \mathbf{8.294} \approx \mathbf{415}$$

Interpretation: We need at least 415 samples to guarantee with 95% confidence that all hypotheses in H will have their in-sample error within ± 0.1 of their true out-of-sample error.

2.6 Putting it all Together …

- **Let's Define a Theorem:**

Theorem - Generalization Bound for Finite Hypothesis Classes

Consider a learning problem with:

- **Dataset** $\mathbf{D} = \{\mathbf{x_1}, \ldots, \mathbf{x_N}\}$
- **Hypothesis class** $\mathbf{H} = \{\mathbf{h_1}, \ldots, \mathbf{h_M}\}$.
- Suppose **g** is a final Hypothesis selected by the learning algorithm.

Then, for any $\varepsilon > 0$, the probability that the generalization error exceeds ε is bounded by:

$$\mathbb{P}\{|E_{\text{in}}(g) - E_{\text{out}}(g)| > \varepsilon\} \le 2Me^{-2\varepsilon^2 N}$$

- $\mathbf{E_{in}(g)}$: In-sample error (training error)
- $\mathbf{E_{out}(g)}$: True out-of-sample error
- $\mathbf{M}$: Size of hypothesis class
- $\mathbf{N}$: Number of training samples

2.6.1 Putting it all Together …

Significance of the Generalization Bound Theorem

Key Messages:

- **Message 1: Bounding the Unknown with the Known**
  - $\mathbf{E}_{\text{in}}(\mathbf{h})$: Training error you *can* compute
  - $\mathbf{E}_{\text{out}}(\mathbf{h})$: True error you *want* to know
  - The theorem guarantees they are close when $\mathbf{N}$ is sufficiently large:

$$|\mathbf{E}_{\text{in}}(\mathbf{h}) - \mathbf{E}_{\text{out}}(\mathbf{h})| \leq \varepsilon$$

    with high probability $(\mathbf{1} - \delta)$

- **Message 2: Universal Guarantee**
  - The bound $\mathbf{2Me^{-2\varepsilon^2 N}}$ is *independent* of:
    - ∗ The specific hypothesis $\mathbf{h}$
    - ∗ The input distribution $\mathbf{p}(\mathbf{x})$
    - ∗ The target function $\mathbf{f}$
    - ∗ The learning algorithm $\mathcal{A}$
  - This makes it a **universal upper bound** that holds for:
    - ∗ Any learning algorithm
    - ∗ Any hypothesis class $\mathbf{H}$
    - ∗ Any data distribution

{Q: Why Deterministic Infeasibility Doesn't Kill Probabilistic Feasibility?}

PAC Framework – Probabilistic Views of Learning.

2.7 Same Reality, Different Lenses: Deterministic vs. Probabilistic Analysis.

- **Deterministic analysis** suggests **that learning is infeasible**,
  - while **probabilistic analysis** indicates **it is feasible**.
- At first glance, this seems contradictory, but a closer look reveals there is actually **no conflict**.
- The **"infeasible vs. feasible"** difference comes from what
  - **each analysis is measuring and the assumptions they make.**
- **Here is the reasoning:**
  1. **Deterministic analysis:**
     - It asks: "**Can we guarantee perfect learning for all possible datasets and noise conditions?"**
     - This is a **worst-case guarantee** – it must hold no matter **how adversarial or unlucky the data** is.
     - Under realistic assumptions (finite data, noisy labels), **the answer is no**,
       - So, **learning appears infeasible**.
  2. **Probabilistic analysis:**
     - It asks: **"What is the likelihood that we learn successfully, given a distribution on the data?"**
     - Here, we **do not require success** for **every possible dataset**,
       - only for **"most" datasets sampled from the distribution**.
     - By **bounding probabilities (e.g. Hoeffding, Chebyshev),**
       - we can show **that with high probability**, **learning succeeds as the sample size grows**.

2.7.1 Same Reality, Different Lenses: Deterministic vs. Probabilistic Analysis.

- **Why they do not Contradict?**
  - They answer different questions:
    - one is a ***worst-case guarantee***, the other a *high-probability guarantee*.
  - **Infeasibility** in the **deterministic sense** simply means
    - **there will always be some pathological cases;**
      - **probabilistic feasibility** means **those cases are rare enough to ignore in practice**.

3. Towards Generalization of Learning.

{ From A PAC Perspective.}

3.1 To Summarize – PAC.

1. Guarantee and Possibility:

Deterministic vs. Probabilistic Generalization

Deterministic Perspective:

- *Question:* Can dataset $\mathbf{D}$ provide certain knowledge about $\mathbf{f}$ outside $\mathbf{D}$?
- *Answer:* No - for any unseen example, uncertainty about $\mathbf{f}$ always remains.

Probabilistic Perspective:

- *Question:* Can $\mathbf{D}$ provide probable knowledge about $\mathbf{f}$ outside $\mathbf{D}$?
- *Answer:* Yes - we can establish statistical guarantees about $\mathbf{f}$'s behavior.

The fundamental difference lies in accepting probabilistic bounds (via generalization theory) versus demanding absolute certainty (which is impossible for unseen data).

3.1.1 To Summarize – PAC.

2. Role of the Distribution:

Data Generation and Generalization Guarantees

The Role of Data Distribution:

- Both training ($\mathbf{D}$) and test samples are drawn *independently* from the *same* underlying distribution $\mathbf{p_X(x)}$
- This i.i.d. (independent and identically distributed) assumption is crucial for:
  - Applying Hoeffding's inequality
  - Ensuring generalization bounds hold
- Test points are *new* (not in $\mathbf{D}$) but come from the same $\mathbf{p_X(x)}$

Implications for Learning:

- *Deterministic Case:* Impossible to guarantee behavior on unseen points
- *Probabilistic Case:* Can bound $|\mathbf{E_{in}(h)} - \mathbf{E_{out}(h)}|$ because:
  - Training and test samples share $\mathbf{p_X(x)}$
  - Bounds hold uniformly across all $\mathbf{h} \in \mathbf{H}$

The i.i.d. assumption bridges the gap between empirical observations (on $\mathbf{D}$) and expected performance (on new data).

3.1.2 To Summarize – PAC.

3. **Learning Goal:**

The Two-Level Approximation Framework for Learning

Learning Goal:

$$\text{Minimize the true error: } \mathbf{E}_{\text{out}}(\mathbf{g}) \approx \mathbf{0}$$

Achieved through two-level approximation:

$$\mathbf{E}_{\text{out}}(\mathbf{g}) \overset{\text{Hoeffding}}{\approx} \mathbf{E}_{\text{in}}(\mathbf{g}) \overset{\text{Training}}{\approx} \mathbf{0}$$

- **First Approximation (Generalization):**
  - Guaranteed by Hoeffding's Inequality
  - Requires sufficiently large $\mathbf{N}$ (sample size)
  - Ensures $\mathbf{E}_{\text{out}}(\mathbf{g}) \approx \mathbf{E}_{\text{in}}(\mathbf{g})$ with high probability
- **Second Approximation (Training):**
  - Depends on hypothesis set $\mathbf{H}$ and learning algorithm $\mathcal{A}$
  - Requires finding $\mathbf{g} \in \mathbf{H}$ with minimal $\mathbf{E}_{\text{in}}(\mathbf{g})$
  - Reflects optimization capability

Implications for Learning Theory:

- The approximation chain connects:
  - Statistical theory (first approximation)
  - Algorithmic performance (second approximation)
- Reveals fundamental trade-offs between:
  - Hypothesis set complexity $|\mathbf{H}|$
  - Sample size $\mathbf{N}$
  - Approximation accuracy

3.1.3 To Summarize – PAC.

4. More complex $\mathcal{H}$:

The Hypothesis Complexity Trade-off

Complexity Dilemma: For hypothesis class $\mathbf{H}$ with size $\mathbf{M}$:

$$\mathbf{E}_{\text{out}}(\mathbf{g}) \overset{\text{Generalization}}{\approx} \mathbf{E}_{\text{in}}(\mathbf{g}) \overset{\text{Training}}{\approx} \mathbf{0}$$

- **When M increases (complex H):**
  - ↑ Looser generalization bound (Hoeffding becomes less reliable)

$$\mathbb{P}\{|\mathbf{E}_{\text{in}}(\mathbf{g}) - \mathbf{E}_{\text{out}}(\mathbf{g})| > \varepsilon\} \leq \mathbf{2Me}^{-\mathbf{2}\varepsilon^{\mathbf{2}}\mathbf{N}}$$

  - ↓ Better training approximation (more candidates to minimize $\mathbf{E}_{\text{in}}$)
- **When M decreases (simple H):**
  - ↓ Tighter generalization guarantee
  - ↑ Risk of large training error

The Fundamental Trade-off:

Complexity	$\longleftrightarrow$	**Sample Size**
Training Accuracy	$\longleftrightarrow$	Generalization Gap

Open Question:

- Where is the optimal balance between:
  - Hypothesis complexity $\mathbf{M}$
  - Training data size $\mathbf{N}$
  - Desired accuracy ε
- Requires analysis of:
  - VC dimension (for infinite H)
  - Regularization techniques
  - Structural risk minimization

3.1.4 To Summarize – PAC.

5. **More complex f:**

Impact of Target Function Complexity

Effect of Complex Target Function f:

$$E_{\text{out}}(g) \overset{\text{Unaffected}}{\approx} E_{\text{in}}(g) \overset{\text{Hard}}{\approx} 0$$

- **First Approximation (Generalization):**
  - Remains valid regardless of f's complexity
  - Hoeffding bound $2Me^{-2\varepsilon^2 N}$ depends only on:
    - Hypothesis class size M
    - Sample size N
    - Tolerance ε
- **Second Approximation (Training):**
  - Becomes challenging for complex f
  - May require larger M to reduce E_{in}
  - But increasing M weakens generalization guarantee

The Complexity Trade-off:

$$\begin{array}{ccc} & \uparrow \text{Complex } f \Rightarrow \text{Harder to achieve } E_{\text{in}} \approx 0 & \\ & \Updownarrow & \\ \text{Solution: Enlarge } H & \Rightarrow & \text{Risk: } E_{\text{in}} \not\approx E_{\text{out}} \end{array}$$

Key Insight:

- Complex f primarily affects training approximation
- Resolution requires careful balance of:
  - Hypothesis class richness
  - Generalization guarantees
  - Available data N

3.2 What if H is infinite?

- **PAC Learning:**
  - **PAC (Probably Approximately Correct) learning** works well when the **hypothesis space H is finite**.
  - **Hoeffding inequality** provides **bounds** for the **difference** between **in-sample and out-of-sample errors**:
    - $P[|E_{in}(h) - E_{out}(h)| > \epsilon] \leq 2|H|e^{-2\epsilon^2 N}$
- **For Finite H in PAC learning:**
  - We could use the bounds like:
    - $N \geq \frac{1}{\epsilon}\left(\log|H| + \log\frac{1}{\delta}\right)$
  - This tells us **how many samples we need to learn H with accuracy ϵ and confidence $1 - \delta$.**
- **Question: What if $|H|$ is infinite?**
  - The bound becomes useless i.e.
    - $\log|H| \rightarrow \infty$, suggests we'd need **infinitely many samples** which **sounds hopeless**.
  - We can **no longer guarantee generalization** from **training data using Hoeffding directly**.

3.2.1 Challenge the "infinite = impossible" idea:

- **Infinite hypothesis class** doesn't necessarily mean **infinite complexity**:
  - we need a **better measure of capacity** than just **|H|**.
  - We need something that works for **both finite and infinite H.**
- **The Core Question:**
  - The need for a **new measure of complexity**:
    - **How doe we measure the capacity or complexity of an infinite hypothesis space?**
    - **How can we ensure that in – sample error still approximates out – of – sample error?**

Next: VC Analysis …

Thank You